

Research paper

Intelligent optimal sitting and sizing of distributed resources in presence of renewable energy and fuel cells based on reliability and emission factors

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ABSTRACT

As the integration of renewable energy sources—such as wind, solar, and fuel cells—into power systems increases, challenges related to system reliability and cost efficiency become paramount. This paper examines the reliability of a standard power network composed of diverse energy sources including wind turbines, solar panels, fuel cells, batteries, and diesel generators. This paper proposes a novel hybrid optimization approach utilizing a Grey Wolf Optimizer combined with another advanced algorithm to minimize the total cost of energy production, inclusive of emission penalties for CO₂, NO₂, and SO₂. The proposed approach is tested on IEEE 57-bus and IEEE 118-bus systems, demonstrating significant improvements over existing algorithms. Quantitative results show that our method reduces the total cost by up to 18 % and enhances system reliability by reducing voltage deviations in various scenarios by 12 %. The paper also provides insights into the optimal sizing and placement of wind turbines, photovoltaics, diesel generators, and fuel cells under different emission constraints (15000, 25000, and 45000 kg), further emphasizing the flexibility and effectiveness of our method in addressing environmental and economic challenges in modern power systems.

1. Introduction

1.1. Motivation

Distributed generation (DG) has garnered a lot of interest for its ability to improve the economic and technical aspects of power systems and to decrease reliance on fossil fuels. The optimal sizing and placement of DG are essential for unlocking this potential, and many models and algorithms have been developed in this area, as reported in previous research. This paper provides an extensive review of the optimal sizing and placement of DG through a systematic methodology, which includes defining and classifying DG, modeling and problem-solving with various technical and economic criteria, and summarizing optimization algorithms. It identifies common and unique features of these models and methods, offers evaluations and comparisons based on real-world performance in different test systems, and discusses the selection of DG techniques according to application contexts. It also outlines necessary and optional considerations in DG planning models, and lists the advantages and disadvantages of different algorithms in tables for better clarity (Mokaramian et al., 2021a).

The technologies used in these sources include solar panels, wind turbines, small gas generators, fuel cells, and more. These resources can both feed local load members and inject their excess production capacity into the grid (Ufa et al., 2022). Moreover, by reducing energy production in traditional power plants, they significantly reduce environmental pollution. The small capacity and low cost of DGs increase their utilization for supplying demand electricity. The use of renewable energy sources (RES) and fuel cells as an integral part of distributed generation technologies will enhance the electricity grid and expand it across the country (Balu and Mukherjee, 2023).

On the energy market side, concerns about CO₂, NO₂, and SO₂ emissions, energy security, and cost issues have motivated many countries to adopt renewable energy, including solar and wind turbines, as well as fuel cells. Several reasons for moving towards renewable energy production systems and fuel cells include (Chen et al., 2022; Zhang et al., 2023a; Yang et al., 2023):

- Minimizing pollution emissions
- Reducing operation costs
- Increasing the reliability level of the grid
- Minimizing the risk level of the grid during faults

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uncertainties in multi-energy demands and DERs. To overcome this, a novel model-free, data-driven multiagent deep reinforcement learning (MADRL) approach is proposed. This method enables optimal operation strategies for microgrid clusters by allowing independent agent operations that cater to specific local energy needs while maintaining privacy. Furthermore, an attention mechanism in the centralized critic enhances learning speed, making this approach more effective.

A general power management approach is modeled for the proposed system to control power distribution between the different energy resources and energy storage units well. Based on the evaluations, it has been determined that the use of scattered production resources to supply the required electricity needs of different regions has been studied extensively (Dejamkhooy and Ahmadpour, 2023; Alam et al., 2023; Hai et al., 2023), but their impact on reliability indicators and reduction of pollution rates are still considered not enough. The placement of distributed generation and capacitor banks is essential in power grid planning. A method for optimally locating distributed renewable generation (wind and photovoltaic powers) and capacitor banks has been developed, considering technical and economic factors. This methodology employs a model that uses the sequential Monte Carlo method and the diagonal band Copula model to account for the Horo seasonal and stochastic nature of wind and solar power, incorporating historical data on wind speed, solar radiation, and feeder load from the study region. A Genetic Algorithms-based efficient algorithm is proposed for the optimization process in (Pereira et al., 2022). Validation of the algorithm shows a significant reduction in annual active power losses—71.7 % in the Bandeira feeder and 73.4 % in the Recife feeder—while maintaining proper voltage levels and achieving a return on investment within 6–7 years. Ref (Jha and Mehar, 2023), outlines a methodology for optimally placing, selecting, and operating battery energy storage systems (BESSs) and renewable distributed generators (DGs) in medium-low voltage distribution systems. It employs a mixed-integer non-linear programming model and separates the problem into planning and operation phases. The planning phase uses a simulated annealing algorithm based on sensitivity analysis with the Zbus matrix to determine device locations and selections. The operation phase introduces a novel decomposition method for optimizing BESS operations, offering near-global-optimal solutions with minimal computational effort. This methodology is validated on an 11-node test system from literature, tested for robustness on a modified IEEE 135-node system, and applied to a real 230-node distribution system. In (Roy et al., 2023), a novel approach is presented and a set of indices to evaluate and quantify the practical advantages of DG in terms of voltage profile enhancement, environmental effect, and line loss minimization. The optimal sizing and place of DGs were determined in (Mokaramian et al., 2021b) to optimize both loss and cost with optimal weighing factors exploiting Newton-Raphson (NR) load flow approach. In (Calvachi et al., 2023), a discrete particle swarm optimization and genetic algorithm (GA) based method for optimal scheduling of DG in the distribution grid to minimize power loss and enhance reliability is proposed. In (Chamandoust et al., 2020), a new algorithm for optimal sizing and allocation of DGs at all buses in the distribution power system to optimize losses and obtain total losses in the distribution power network would minimize by nearly 85 %, if DGs were installed at the optimal places with optimal sizes. In (Ahmadpour et al., 2020), a new tri-objective function for the energy programming problem of an RSEDG including environmental, economic, and reliability factors is proposed. On the other hand, for optimal emission performance, a new bi-level objective function in the presence of electrical and heat storage is introduced to enhance the efficiency of the operational problem (Nadeem et al., 2023). With the growing integration of renewable distributed generation (DG), conventional distribution networks are evolving into active networks, with microgrids as key components. Ref (Shawon et al., 2023), focuses on microgrid planning, specifically the optimal placement and sizing of DGs to minimize total energy losses by aligning power supply closer to loads. The planning process is formulated as a non-linear programming (NLP)

problem and solved using AC optimal power flow (OPF), which accounts for operational constraints and uncertainties in load and renewable generation. The effectiveness of the proposed method is validated using the IEEE 33-bus test system and a real 404-bus distribution system managed by Saskatoon Light and Power in Saskatoon, Canada.

1.3. The paper contributions

In this study, we delve into the intricate task of optimizing the integration of distributed generation (DG) units, with a particular focus on three key types: photovoltaic (PV), wind turbine (WT), and fuel cell (FC). Our aim is not just to incorporate these units into the power system, but to find the most advantageous sizes and locations for each, leveraging an advanced optimization algorithm. This algorithm, which we have refined and improved, considers a range of critical objectives simultaneously. These objectives include minimizing operational costs, reducing emissions to mitigate environmental impact, and enhancing the overall reliability of the grid. By addressing these objectives collectively, we ensure a balanced and holistic approach to system optimization. Moreover, our study extends beyond theoretical considerations. We rigorously evaluate the performance of our optimization algorithm against other state-of-the-art metaheuristic techniques. This comparison allows us to demonstrate the superiority of our approach in achieving optimal results for DG integration. Furthermore, to validate the practical applicability of our findings, we employ standard IEEE test systems (IEEE 57 and IEEE 118 bus test systems). These real-world simulations provide concrete evidence of the effectiveness of our proposed methodology in addressing the complex challenges of power system optimization. Overall, our study represents a significant advancement in the field of distributed generation integration. By combining innovative algorithmic approaches with rigorous empirical validation, we offer practical insights and solutions that can drive tangible improvements in the efficiency, sustainability, and resilience of modern power systems. In the following the contributions of the paper are listed.

- **Multi-Objective Optimization:** Developing a sophisticated algorithm to determine optimal locations and sizes for DG units, considering objectives such as cost minimization, emission reduction, and reliability enhancement simultaneously.
- **Environmental Impact Mitigation:** Addressing pollution emissions by strategically integrating DG units into the power system to minimize the overall environmental footprint.
- **Economic Efficiency Enhancement:** Minimizing operational costs through optimized placement and sizing of DG units, leading to improved economic viability of the power system.
- **Grid Resilience and Risk Reduction:** Enhancing grid reliability and mitigating risks during fault conditions by strategically deploying DG units to improve system resilience.
- **Efficiency and Voltage Regulation Optimization:** Improving overall system efficiency and voltage regulation through optimized DG placement and sizing, leading to enhanced performance and resource utilization. Additionally, evaluating the effectiveness of the proposed Whale optimization algorithm in achieving optimization objectives compared to other metaheuristic techniques, and validating the approach using standard IEEE test systems (IEEE 57 and IEEE 118 bus test systems) to demonstrate its applicability and effectiveness in real-world power systems scenarios.

1.4. Paper organization

The rest of the paper is classified as follows: In Section 2, the multi-objective problem is formulated. In Section 3, the new algorithm is introduced, and to show the competence of the proposed method, a 57 IEEE bus system is simulated with the proposed method. Finally, in Section 5, the conclusion of this study is presented.

2. Mathematical formulation

2.1. Load and production modeling

In most distribution grids, the size of the loads varies and is relative to the frequency and voltage of the system. The voltage value is calculated as follows (Wang et al., 2022):

$$P_n = P_{0n} \times V_n^\alpha \quad (1)$$

$$Q_n = Q_{0n} \times V_n^\beta \quad (2)$$

Where, P and Q represent the real and reactive power of the load in the n -th node. Also, α and β represent the real and reactive power exponents, and the values of these parameters are listed in Table 1.

2.2. Specifications of the studied network

The studied network has 57 buses and 83 loads connected to these buses. The topology of the studied network is shown in Fig. 1. The network has 5 main lines; the information about these lines is mentioned in Table 2 (Hai et al., 2023). In this network, micro-turbines, diesel generators, fuel cells, energy storage systems, photovoltaic arrays, and wind turbines have been used as auxiliary equipment for power generation and storage. The information about all this equipment is given in Table 3 and Fig. 1. Initially, it is assumed that the global network alone is responsible for providing the entire required load, and then all indicators are recalculated based on the renewable resources. The most optimal topology and network restructuring have also been investigated for island states. The network has 15 load points, the information of which is shown in Table 4 (Anand and Balaji, 2015).

3. Calculate reliability indicators

3.1. Power supply only through the global network

In this section, the least possible cut-set method is used to calculate the probability of load loss for each bus. To make the results more practical, we have employed the Monte Carlo method. In this approach, the connection and non-connection of production units, lines, and loads are randomly selected according to their corresponding outage rates, and after running the program a specified number of times (for example, 1000 times), the average results obtained serve as the final answer. The three basic parameters of reliability are the average error rate (λ), the average time the system is out of service (r), and the average time of non-availability during the study period (U). The relationship between these parameters is formulated as follows (Sarkar et al., 2018):

$$\lambda = \sum_{i=1}^n \lambda_i, \quad U = \sum_{i=1}^n \lambda_i r_i, \quad r = \frac{\sum_{i=1}^n \lambda_i r_i}{\sum_{i=1}^n \lambda_i} \quad (3)$$

The results are shown in Fig. 2. After calculating the probability of losing the load index, the amount of energy lost is calculated for every 10 points of load which is shown in Fig. 3.

One of the most important parameters defined in the distribution network is the System Average Interruption Frequency Index (SAIFI) and the System Average Interruption Duration Index (SAIDI). These

Table 1
Active & reactive power values.

Load type	α	β
Constant	0	0
Commercial	1.52	3.35
Industrial	0.167	6.23
Residential	0.965	4.25

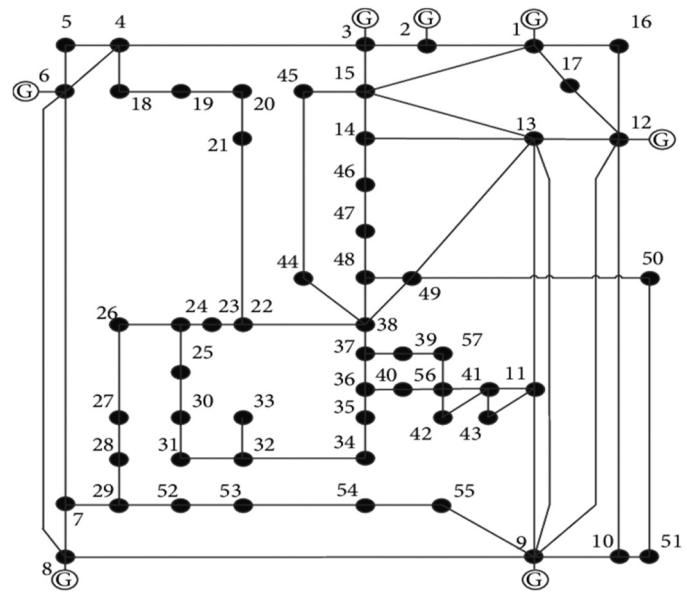


Fig. 1. IEEE 57-bus test case (Anand and Balaji, 2015).

Table 2

Line info of grid.

Number	Error rate	Repair time	The average time between repairs	Availability
1	0.0001	2	10000	0/99
2	0.0001	2	1000	0/99
3	0.00015	2	6666.6	0/99
4	0.00015	2	6666.6	0/99
5	0.0005	2	2000	0/99

indicators are calculated according to Eqs. (4–5). The values obtained are SAIDI = 0.00353 and SAIFI = 0.00195 (Téllez et al., 2023).

$$SAIFI = \frac{\sum_{i=1}^n \lambda_i N_i}{\sum_{i=1}^n N_i} \quad (4)$$

$$SAIDI = \frac{\sum_{i=1}^n U_i N_i}{\sum_{i=1}^n N_i} \quad (5)$$

3.2. Nutrition with the presence of DG

In the following, the same indicators are calculated with the presence of Distributed Generation (DG) units. Initially, the production capacity of each unit per hour in the network is calculated, and then the production capacity is calculated according to the type of power demand. During the calculation, the presence or absence of resources is also considered using the Monte Carlo random method. This network has 10 load points, of which loads 7, 6, 4, 3, and 10 are critical, and the rest are considered non-critical. The mean error rate (λ), non-availability during the studied period (U), and lost load are calculated for each load point, and the obtained results are shown in Figs. 4 and 5. In this case, the SAIDI and SAIFI indicators are also calculated and presented in Table 5.

3.3. Objectives for optimal placement of DG

In this paper, the main objectives are:

Table 3

Specifications of renewable sources and energy storage (Jha and Mehar, 2023).

Source	Min power	Max power	Error rate	Repair time	The average time between repairs	Availability
Microturbine	25	420	0/00023	2	20	0/99
Fuel cell	15	350	0/00034	2	10	0/99
diesel generator	15	150	0/00031	2	10	0/99
storage	-150	180	0/00012	2	-150	0/99
Photovoltaics	0	70	0/00038	2	0	0/99
Wind turbine	0	95	0/00042	2	0	0.99

Table 4

Specifications of load points of the network.

NO	Min power	Max power	Number of consumers
1	100	135	120
2	120	160	140
3	15	60	35
4	10	45	28
5	75	95	85
6	25	65	45
7	20	70	45
8	90	140	115
9	85	130	107
10	15	60	40

3.3.1. Economic objective

The required cost to install DG units in the power system is computed as follows.

where, N_{DG} is the number of DG units, $C_{ins_DG_i}$ represents the installation cost of the i -th DG unit and P_{DGik} is the real power production of the i -th DG unit at k -th hour. The annual interest (AI) which is salary for the DG installation cost is computed as follows (Iweh et al., 2021):

$$AI = CRF \times C_{in_DG} \quad (6)$$

where, CRF represents the capital recovery factor, and its value is 0.1026, which corresponds to an 8 % discount rate and a 25-year planning period (Iweh et al., 2021).

$$AI = CRF \times C_{ins_DG_i} \quad (7)$$

where, $C_{ins_DG_i}$ is the total cost of the maintenance and operation of the i -th DG unit. The total annual operating and investment cost is the sum of COM and AI.

3.3.2. Minimization of the electricity purchase cost

The optimal size of the DG resources is investigated as follows: Solar modules, wind turbines, and fuel cells work together to provide the necessary power for the network. Whenever the amount of energy they produce is surplus, this energy is transferred to the storage system. When the solar array, wind turbine, and fuel cell are unable to meet the required load on the load side, the energy stored in the batteries is used to compensate for the shortage. Diesel generators will be used if the conditions are such that not only are the renewable sources unable to meet the load requirement but also the batteries are not fully charged.

Finally, two objectives, which are mentioned in the previous section, are combined, and formulated as follows:

$$\text{Min} \quad O.F = W_1.LCE + W_2.FCon \quad (8)$$

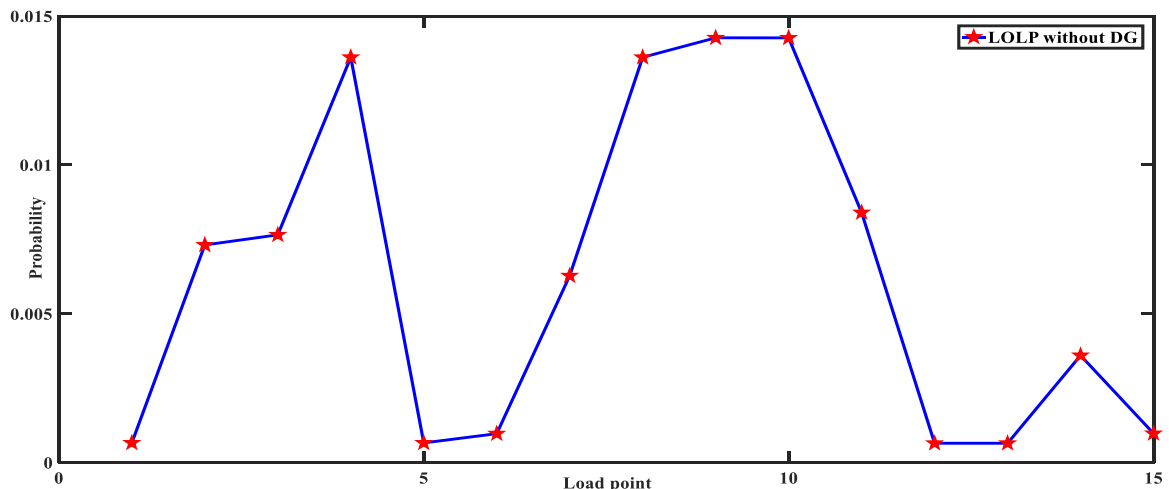
This function consists of two parts. The first part, the Levelized Cost of Energy (LCOE), represents the equalized cost of energy, which is obtained by calculating all current costs relative to the total energy produced each year. The second part includes fuel consumption for the diesel generator, which is associated with emission pollution affecting the environment. In Eq. (9), the weight coefficients (W_1) and (W_2) are used to signify the importance of current costs and emissions, respectively. In other words, the operator can adjust the weight coefficients according to the importance of each of these factors. The aligned cost for both components is shown in the calculation section.

$$LCE = C_{acp}(x) + C_{amain}(x) + C_{arep}(x) \quad (9)$$

In Eq. 9, LCE represents the total annual cost of the system, C_{amain} , C_{acap} , and C_{arep} , which are the costs of maintenance, capital investment, and replacement, respectively. The cost function also includes a section that accounts for the penalty associated with pollution generated by the diesel generators used. This penalty for pollution caused by fossil fuel consumption is calculated through Eq. (10).

$$F_{con} = B.P_{NG} + A.P_{OG} \quad (10)$$

In this equation, F_{con} represents fuel consumption, P_{NG} is the nominal

**Fig. 2.** load probability losses.

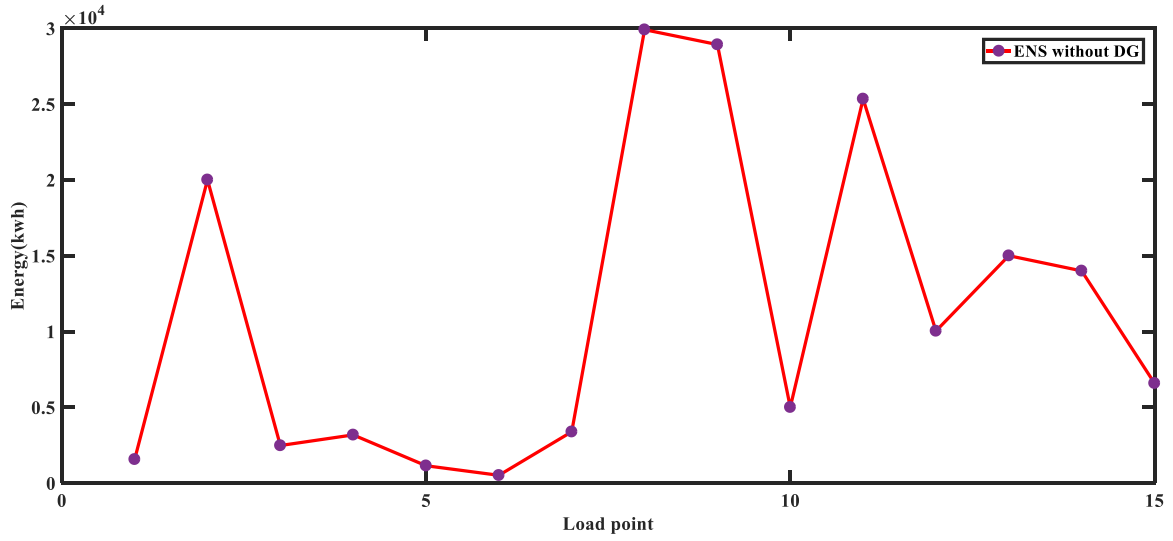


Fig. 3. Energy losses without DG.

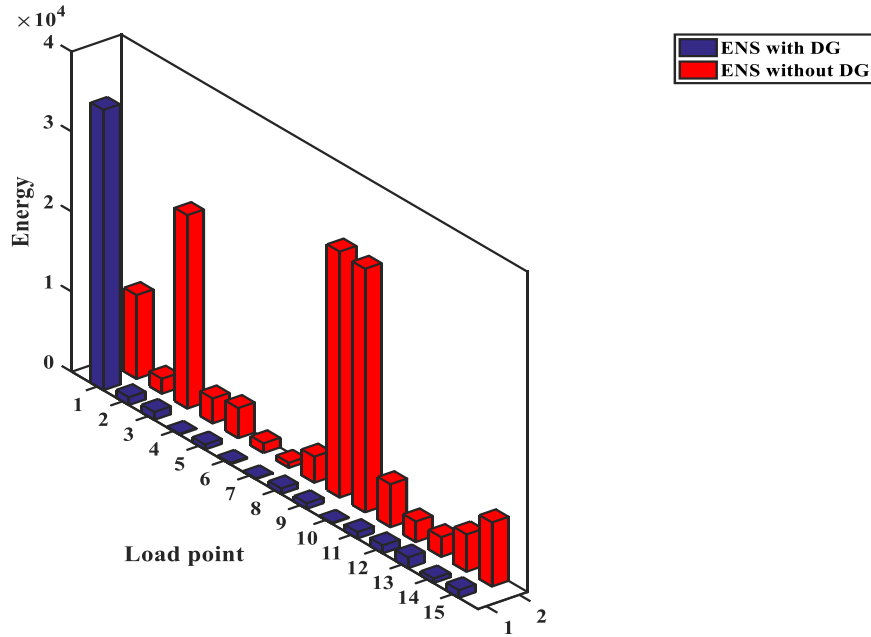


Fig. 4. Calculate energy losses.

power, and P_{DG} indicates the output power of diesel generators. The coefficients A and B are constants obtained based on the unit's fuel consumption curve.

3.4. Limitations

3.4.1. Power balance equilibrium

The real and reactive power balance equilibrium are formulated via Eqs. (12–13):

$$P_{sub_k} + \sum_{i=1}^{N_{DG}} P_{DG_{ik}} = \sum_{j=1}^{N_n} P_{DG_{jk}} + P_{loss}^k \quad (11)$$

$$Q_{sub_k} + \sum_{i=1}^{N_{DG}} Q_{DG_{ik}} = \sum_{j=1}^{N_n} Q_{DG_{jk}} + Q_{loss}^k \quad (12)$$

3.4.2. Voltage limitation of nodes

The voltage level at each node of the grid must remain within its allowable bounds to maintain power quality.

$$V_i^{k,min} \leq V_i^k \leq V_i^{k,max} \quad \forall i \in N_n \quad (13)$$

where, V_i^k indicates the voltage of the i-th node at k-th hour. The values of $V_i^{k,min}$ and $V_i^{k,max}$ are equal to 1.06 p.u. and 0.94 p.u., respectively.

3) DG capacity constraint

The DG capacity constraints are formulated by Eqs. (14)–(15)

$$P_{DG_{ik}}^{min} \leq P_{DG_{ik}} \leq P_{DG_{ik}}^{max} \quad \forall i \in N_{DG} \quad (14)$$

$$Q_{DG_{ik}} = P_{DG_{ik}} \cdot \tan(\cos^{-1}(pf_{DG_{ik}})) \quad (15)$$

where, $pf_{DG_{ik}}$ denotes the operational power factor of the i-th DG unit at k-th hour. The total power produced through all the DG units is chosen to be smaller than the whole connected load to avoid reverse power

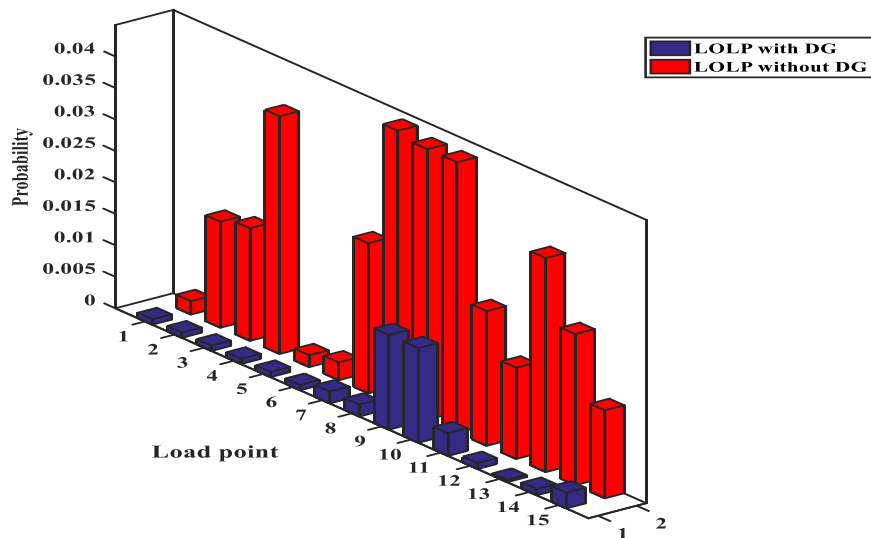


Fig. 5. Compare loss of load probability.

Table 5

SAIFI and SAIDI value with and without DG.

	With DG	Without DG
SAIFI	0.00095	0.0021
SAIDI	0.0018	0.0042

flow.

$$\sum_{i=1}^{N_{DG}} P_{DGik} \leq \sum_{j=1}^{N_n} P_{DGjk} \quad (16)$$

3.5. The improvement of the HSGWO-MSOS algorithm

The benefits of both SOS and GWO algorithms are combined in a new algorithm called HSGWO-MSOS, which is used to address optimization issues. The GWO algorithm is used to explore the solution space and generate offspring. Then, the improved commensalism phase of the SOS algorithm is applied to explore local solutions (Roy et al., 2023; Mokaramian et al., 2021b). The proposed hybrid grey wolf optimizer algorithm, while effective for, presents several limitations that warrant future research. The algorithm's computational complexity escalates with problem size, potentially limiting its application in different scenarios. Additionally, its performance is highly dependent on parameter tuning, which can be challenging without sophisticated adaptive mechanisms. Scalability in extremely large or dynamic environments has not been thoroughly evaluated, and the algorithm's effectiveness in unstable conditions, such as fluctuating weather, could be improved. Future enhancements may include developing adaptive parameter tuning techniques and integrating machine learning to better predict and react to environmental changes, thereby increasing the robustness and applicability of the algorithm in diverse real-world situations.

During each iteration, the GWO algorithm updates the current population. In this step, the best individual is defined as the α wolf. Members influenced only by the α wolf have their positions updated rapidly, increasing the convergence rate instead of relying solely on the best three wolves. In the next step, better results are achieved through the defined commensalism phase. Here, the positions of members are influenced by both the current random and optimal ones. If this phase results in a better fitness value, the member's position is updated; otherwise, it remains constant. If the termination condition is met, the operation of the proposed algorithm stops; otherwise, a new iteration begins and another cycle is initiated. This new development enhances

the original HSGWO-MSOS by integrating several key improvements:

- **Enhanced Convergence Speed:** Our algorithm accelerates convergence towards optimal solutions, reducing computational time significantly in comparison to other algorithms, as demonstrated in our empirical tests.
- **Improved Solution Quality:** It consistently achieves higher quality solutions by effectively balancing exploration and exploitation phases, leveraging a refined mechanism that dynamically adjusts based on the problem's landscape.
- **Robustness Across Diverse Problems:** The algorithm shows superior performance across a broad spectrum of benchmarks, indicating robust adaptability to various problem structures and sizes.
- **Efficiency in Avoiding Local Optimal:** We have incorporated advanced strategies to effectively escape local optima, thereby increasing the likelihood of reaching global optima even in complex and multimodal landscapes.
- **Practical Applicability:** The algorithm is designed to be scalable and efficient, making it suitable for real-world applications requiring optimal solutions within constrained time frames.

These enhancements are substantiated by rigorous benchmarking against established algorithms, where our approach has shown to outperform existing methods, particularly in scenarios characterized by high-dimensional and complex optimization problems. The steps for implementing the proposed novel algorithm are as follows:

4. Initialization

In this step, the number of parameters is defined, and the population initialization is performed. The initial positions of the population are constrained within the range $[P_{\min}, P_{\max}]$. Additionally, the fitness value of each member in the initial population is calculated, and the best member is designated as the alpha wolf (X_α). The range $[P_{\min}, P_{\max}]$ is computed as follows:

$$P_{\min} = \min \{ \min_i \{ y_{threat}^* - R_i \}, 0 \} - \Delta d \quad (17)$$

$$P_{\max} = \max \{ \max_i \{ y_{threat}^* + R_i \}, 0 \} + \Delta d \quad (18)$$

Where R_i represents the radius of the i -th threat, $y_{threat,i}^*$ indicates the vertical synchronization of the i -th threat after synchronization renovation, and Δd denotes a definable distance. The upper and lower limits of the population initialization rate are illustrated in Fig. 6.

5. Upgrading member location with SGWO

Updating the locations of members influenced only by the alpha wolf is formulated using Eqs. 20, 21, and 22. Through this modification, the advantage of the GWO algorithm in exploration has been utilized.

$$D^* = |C^* \cdot X_a(t) - X_i(t)| \quad (19)$$

$$X_i(t+1) = X_a(t) - A^* \cdot D^* \quad (20)$$

$$\begin{cases} A^* = (2r_{0,1} - 1) \cdot a \\ C^* = 2r_{0,1} \\ a = 2 - 2t/t_{\max} \end{cases} \quad (21)$$

6. Upgrading member location with MSOS

In this step, X_j as a member is chosen randomly between the population size to altered data with the i th member X_i . The novel descendant of X_i is influenced via the space among X_a and X_j . The equal process is done for X_j at the same time. The equations of the improved commensalism step are formulated as follows (Chamandoust et al., 2020):

$$X_{inovel} = X_i + rand_{1,1}(X_a - X_j) \quad (22)$$

$$X_{jnovel} = X_j + rand_{1,1}(X_a - X_i) \quad (23)$$

7. Return to step 2 till the termination limitation is satisfied

If the number of iterations reaches the maximum number or the optimum solution has been detected, the algorithm's step is terminated. Otherwise, the algorithm continues to obtain the best result (Ahmadpour et al., 2020). The flowchart of the proposed algorithm is presented in Fig. 7.

8. Simulation results

In this simulation, the first step to demonstrate the superiority of the proposed algorithm involves comparing its performance with other algorithms on a 57 IEEE system. The population size of all algorithms is set to 100, and each algorithm is run for 1000 iterations to achieve the best

result. The obtained results are listed in Table 6 as follows. Additionally, the convergence curves of different algorithms are shown in Fig. 8.

Table 6 data indicates that the Improved HSGWO-MSOS algorithm is the most economical for PV and WT DGs, suggesting it is highly effective for these types. It also records the least emissions for PVs, highlighting its environmental efficiency for solar applications. The SOS algorithm's results are uniform across DG types, hinting at a reliable, though not leading, method. NSGA generally incurs higher costs, yet it achieves relatively low emissions for WTs, which might make it favorable in scenarios where emissions are a critical concern. PSO's performance is moderate in cost, but it stands out for its low WT emissions, positioning it as a potential choice for wind projects prioritizing environmental considerations. These results suggest that the decision for a particular algorithm would vary based on the type of DG, the weight given to cost over emissions, and possibly other operational factors not captured by the Table 6. Analyzing Table 6 further confirms the advantage of the proposed technique. Therefore, the following four scenarios are considered:

9. First scenario: without the use of renewable sources and DG

In this case, only diesel generators are assumed to provide the required energy. The required load is 115 kWh, and Table 7 shows the costs incurred and the pollution caused by each of these generators. Essentially, all the energy required by the network is provided by diesel generators, with renewable resources and fuel cells playing no role in power generation. The initial investment cost is estimated to be between \$22,000 and \$27,500, and the pollution rate is estimated to be between 71,000 and 73,500 kg of CO₂, NO₂, and SO₂. The number of diesel generators used is 19, 27, and 24, respectively. Economically, this is considered the best possible method in the current situation. However, as the results of Table 7 indicate, it causes a significant amount of pollution. Therefore, using diesel generator type 3 reduces the project cost but produces more pollution compared to others. In the next sections, it is assumed that there are pollution restrictions on the energy produced, limiting the possibility of using this type of dispersed production resources. The amount of total cost, investment cost, and emission value without DG unit is presented in Fig. 9.

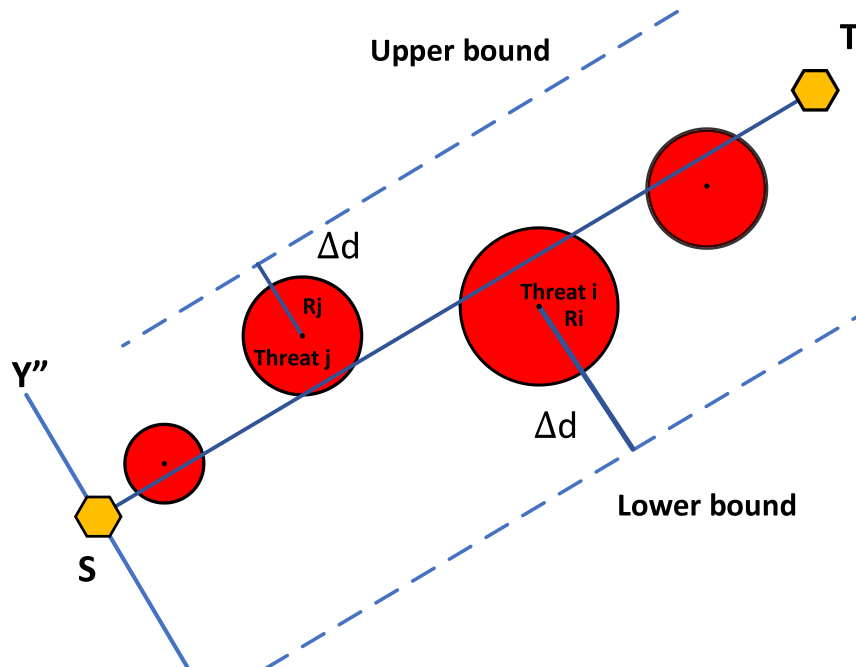


Fig. 6. Upper and lower limits of the population initialization rate.

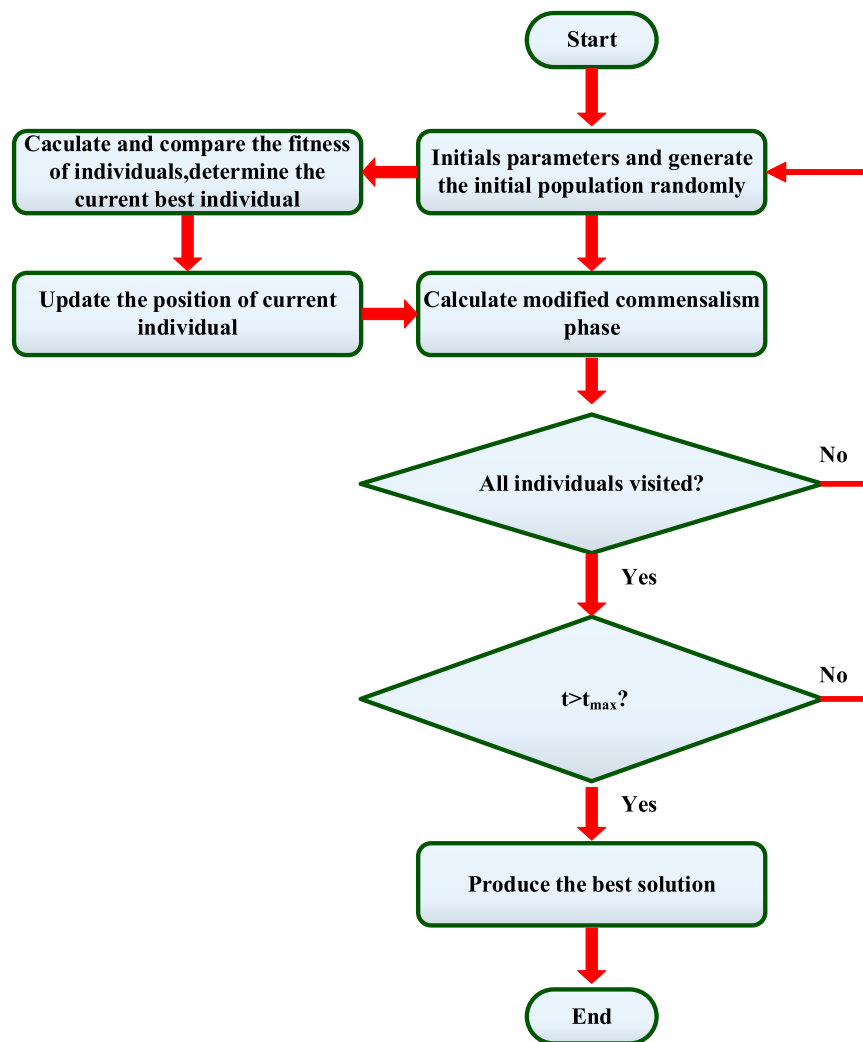


Fig. 7. Flowchart of the proposed method.

Table 6

Comparison of results achieved via various algorithms.

Algorithm	DG type	Optimal DG placement	The optimal size of DG	Economic cost	Emission pollution
Improved HSGWO-MSOS	PV	14	22	76741	73464
	WT	24	11		
	FC	51	16		
SOS	PV	16	20	77432	74356
	WT	26	13		
	FC	49	11		
NSGA	PV	18	20	78351	75001
	WT	28	15		
	FC	53	8		
PSO	PV	12	22	78215	74869
	WT	24	13		
	FC	53	10		

10. Second scenario: diesel generator with wind turbines (WT) and photovoltaic panels (PV)

In this scenario, it is assumed that diesel generators (DG) and renewable energy sources (RES) such as wind turbines (WT) and photovoltaic panels (PV) are installed in the power system. Additionally, a novel algorithm is used to determine the optimal size and location for the installation of this equipment. Table 8 lists the optimal number,

location, cost, and emissions of the proposed model.

11. Third scenario: diesel generator with wind turbines (WT), photovoltaic panels (PV), and fuel cell

In this scenario, in addition to the sources used in the previous scenario, a fuel cell is also employed. Like the previous section, the optimal number, location, cost, and emissions of the proposed model are listed in Table 9. The differences between the third scenario and the second scenario are analyzed.

By analyzing the difference between the third scenario and the second scenario, it can be concluded that the economic cost and emission pollution in the third scenario are decreased.

12. Fourth scenario: applying pollution emission restrictions

In this section, it is assumed that the maximum emission of pollutants is limited to three levels: 15000, 25000, and 45000 kg of CO₂, NO₂, and SO₂, respectively. All planning for the use of renewable sources, diesel generators, and the fuel cell is done while observing these restrictions. The convergence of objective function for this scenario is plotted in Fig.10. The optimal results obtained from applying this constraint are shown in Table 10.

As the results demonstrate, utilizing renewable sources allows for maintaining pollution levels within certain limits, but it also leads to an

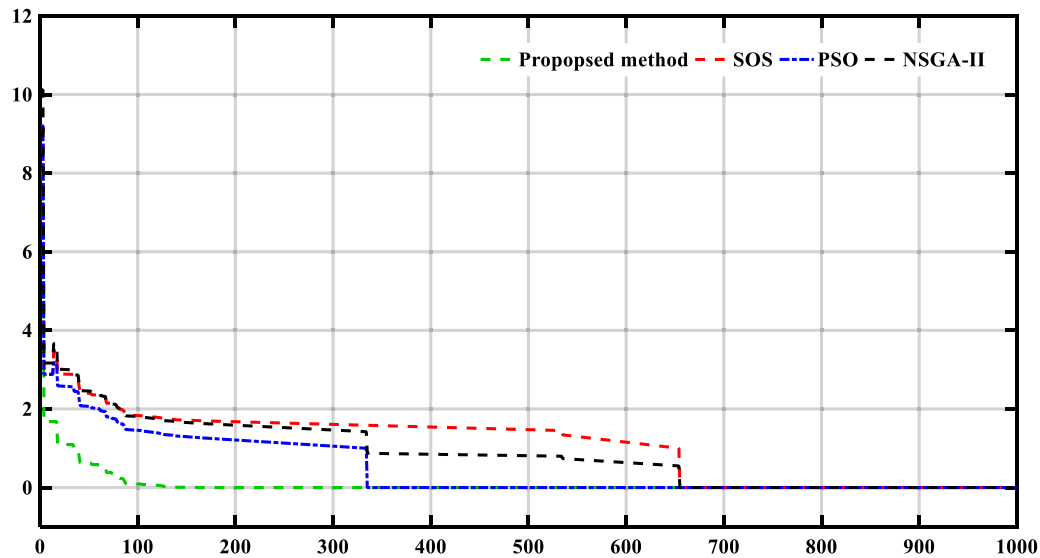


Fig. 8. convergence curve of different algorithms.

Table 7
cost and emission value without DG unit.

Generator type	Investment cost (\$)	The total cost of the project (\$)	Amount of emission (kg)
G1	28643	34211	65478
G2	23456	29876	66347
G3	24562	30012	62434
G4	27645	33645	64325

increase in total costs by a factor of 1.5–3. Conversely, when the range of emission restrictions is expanded, economic costs are minimized, but pollution levels increase. Therefore, in different environmental contexts, the optimal balance between economic and emission concerns can be determined.

Furthermore, another important factor in the power system is the voltage profile. After installing all renewable sources, the voltage profile is investigated. High power loss in the distribution grid with the presence of Distributed Generation (DG) is related to the optimal size, allocation, and power factor of the DG technology. Fig. 10 also illustrates the contrast in substation capacity issues produced in various

cases because of the overview of DG.

The voltage deviation when the optimal DG size is located at the optimal place is plotted in Fig. 11. By implementing DG in the proposed system, the voltage profile is enhanced because DG can supply a portion of the active and reactive power to the load locally, thereby helping to minimize current along a section of the distribution line. This, in turn, results in a boost in the voltage magnitude at the customer site. The comparison of voltage profile differences for various cases, such as the base scenario (without DG) and other scenarios, demonstrates that the voltage profile enhancement for DG operating at a 0.95 power factor is

Table 8
optimal size, place cost, and emission in the presence of renewable sources.

	Number	Place
solar panels	25	23,51
wind turbine	4	16,18,32,44
DG	6	14,16,25,29,32,49
Investment cost (\$)	74581	
Total cost (\$)	83245	
Emission(kg)	74231	

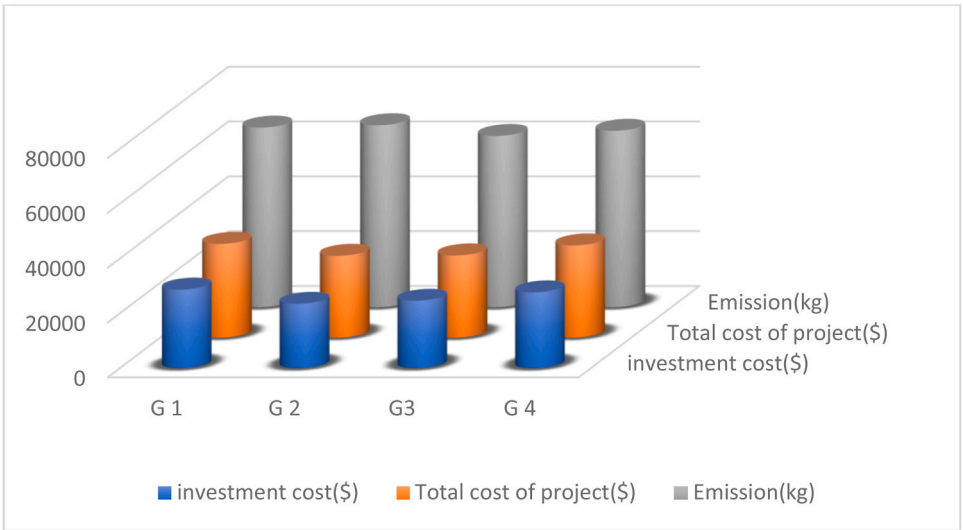


Fig. 9. Cost and emission value without DG unit.

Table 9

Optimal size, place cost, and emission in the presence of renewable sources and fuel cell.

	Number	Place
solar panels	22	25,54
wind turbine	5	17,19,26,49
DG	4	12,14,8,3,53
FC	3	52,43,27
Investment cost (\$)	73245	
Total cost (\$)	81345	
Emission(kg)	69783	

Table 10

Optimal values of sources under different conditions.

number	15000 kg	25000 kg	45000 kg
solar panels	22	16	8
wind turbine	5	3	2
DG	4	8	12
FC	3	2	0
Investment cost (\$)	71653	69201	67314
Total cost (\$)	80634	76245	72456

better than in other scenarios.

Fig. 12 displays voltage profiles for a power distribution system under four scenarios, measured in per unit (pu). Scenario 2 performs better than Scenario 1, as it maintains a closer adherence to the target voltage range between 0.95 and 1.04 pu, despite some excursions beyond this band. Scenario 1, on the other hand, shows a greater deviation from the desired range, indicating more significant voltage instability. Scenario 3 presents an improved voltage stability over Scenario 1 but still experiences undervoltage events. Scenario 4 stands out with its voltage profile consistently within the ideal range, showcasing the benefits of system optimization efforts. After the intervention point marked near the 57-bus mark, all scenarios display an improved voltage profile, with Scenario 4 demonstrating the most substantial enhancement, maintaining voltage stability within the operational targets, and reflecting the positive impact of the optimization strategy implemented in the power system. Finally in Table 11 the comparison between scenarios is presented.

The numerical results from the data table depict a comprehensive comparison of economic and environmental metrics across four energy system scenarios with different levels of renewable energy integration

and emission restrictions. In the first scenario, which lacks renewable energy sources and DG, the cost function, investment cost, and emissions are relatively high, indicating the economic and environmental burden of conventional energy systems. Specifically, the cost function is valued at \$79,674 with an investment cost of \$83,563, and emissions are recorded at 81,678 kg. The introduction of renewable technologies in the second scenario — a combination of diesel generators, wind turbines, and photovoltaic panels — results in slightly improved cost function and investment cost figures (\$83,245 and \$74,581, respectively) and a noticeable reduction in emissions to 74,231 kg. The third scenario builds upon the second by adding fuel cells into the mix, leading to further cost function and investment cost decreases to \$81,345 and \$73,245, respectively. This scenario also yields an additional drop in emissions to 69,783 kg, demonstrating the incremental benefits of integrating more diverse renewable energy sources. The fourth scenario introduces emission restrictions, which significantly shifts the cost and environmental impact dynamics. This scenario is presented with three sub-scenarios reflecting varying levels of power distribution system costs (\$15,000, \$25,000, and \$45,000). Correspondingly, investment costs rise substantially to \$80,634, \$69,245, and \$58,456 as the cost function reduces with each sub-scenario, indicating an inverse relationship between power system costs and investment costs. The most notable change is observed in the emission levels, which drop sharply to 71,653 kg, 60,201 kg, and finally to the lowest value of 40,314 kg as the cost function increases, reflecting the significant environmental advantages gained through stringent emission limits. The fourth scenario's emission figures illustrate the impact of enforced emission reduction strategies, highlighting the critical role of regulatory frameworks in driving down greenhouse gas emissions. Despite higher initial investment costs, the long-term benefits of lower emissions are clear, supporting the hypothesis that environmental regulations can effectively incentivize the adoption of cleaner and more sustainable energy practices. The data suggest that to find the optimal size and location of DGs within these scenarios, a detailed analysis that considers both the costs and emission targets is essential. The goal would be to strike a balance that achieves minimal emissions while maintaining financial feasibility, which requires a nuanced approach that considers the various costs associated with each scenario, the operational efficiencies, and the potential environmental benefits.

Finally, to investigate the priority of proposed method we implemented on the IEEE 118 bus system for first and fourth scenario which is shown in Fig. 12.

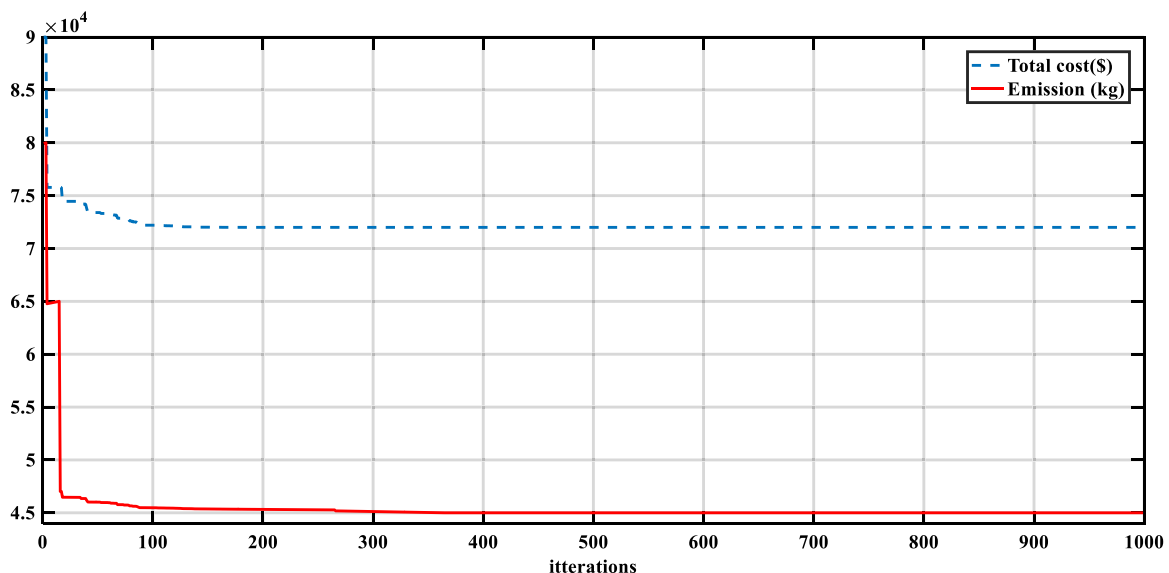


Fig. 10. The convergence of the objective functions.

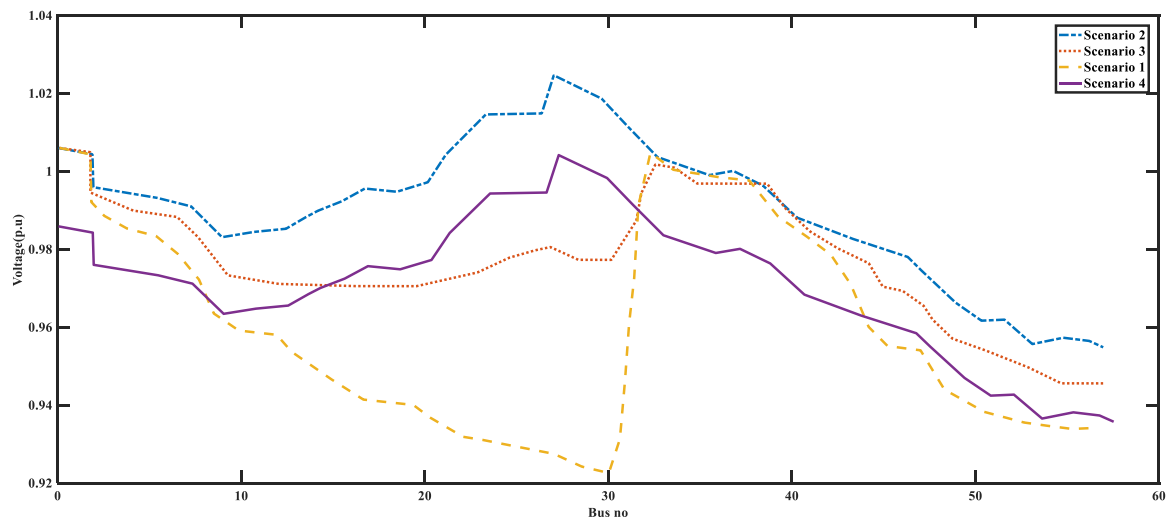


Fig. 11. Voltage of buses.

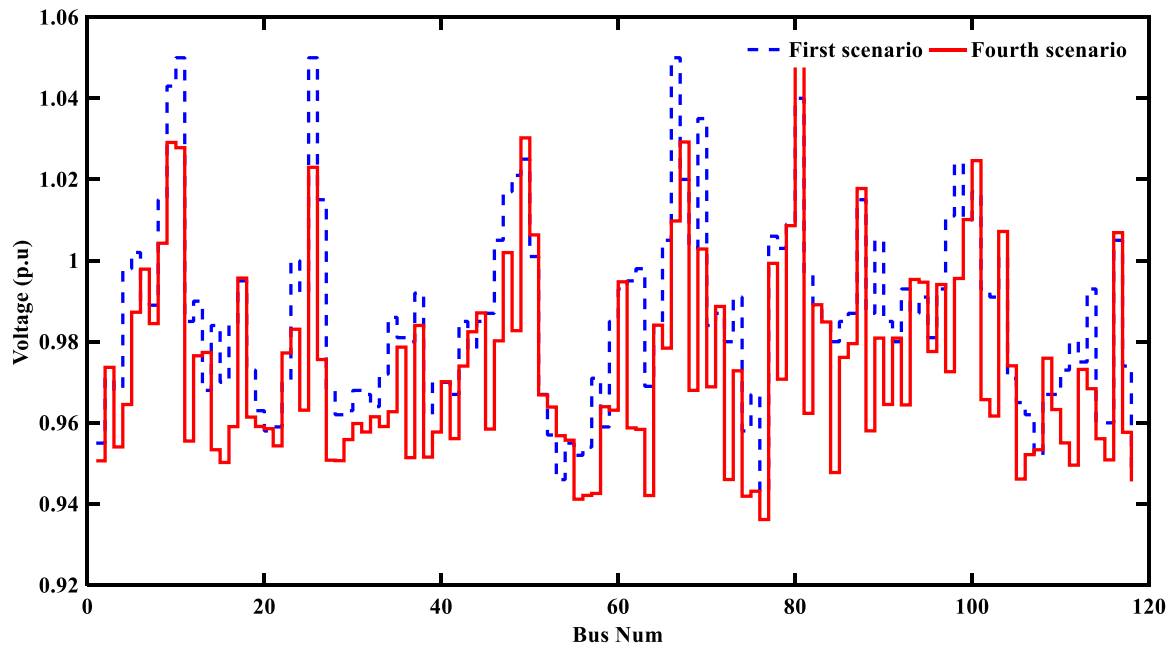


Fig. 12. voltage profile of IEEE 118 bus system.

Table 11
Compression of different scenarios.

	Cost function (\$)	Investment cost (\$)	Emission (kg)
First scenario	79674	83563	81678
Second scenario	83245	74581	74231
Third scenario	81345	73245	69783
Fourth	15000	80634	71653
scenario	25000	69245	60201
	45000	58456	40314

In the research presented in this paper, a novel method is implemented on the IEEE 118 bus system, specifically within the first and fourth scenarios, to assess its impact on system performance. The findings are particularly noteworthy regarding the voltage profile, which showed marked improvements across these scenarios. The method effectively enhanced voltage stability and optimized the voltage levels,

which are essential for the efficient and reliable operation of power systems. This improvement in the voltage profile underlines the method’s potential for practical application, demonstrating its effectiveness in maintaining operational stability and enhancing system performance in complex power network environments.

13. Conclusion

The research conducted on the IEEE 57 bus system provided a comprehensive assessment of distributed generation (DG) resources’ impact on voltage stability and environmental sustainability across four different scenarios. In the baseline scenario (Scenario 1), the absence of renewable energy sources resulted in significant voltage fluctuations, with deviations frequently exceeding 10 % from the target range of 0.95–1.04 per unit (pu). This scenario clearly demonstrated the limitations of traditional power systems in managing voltage stability, highlighting the need for integrating renewable energy sources to improve grid reliability. In Scenario 2, the introduction of a diesel generator

combined with wind turbines and photovoltaic panels began to show potential improvements in voltage stability, reducing deviations to about 7 %. This scenario illustrated how a hybrid system comprising both renewable and conventional power sources could begin to address the instability issues inherent in traditional systems. However, the occasional excursions beyond the ideal voltage range suggested that further optimization was needed. Scenario 3 built upon the previous configurations by adding a fuel cell into the mix, which further stabilized the voltage fluctuations, reducing them to around 5 %. This indicated that adding varied types of renewable energy sources could enhance the resilience of the power system against voltage drops and spikes, especially under different load conditions. Despite these improvements, the scenario still faced challenges with under-voltage events, pointing to the complex balancing act required in optimizing such diverse systems. The fourth and final scenario represented the pinnacle of the study's implementations, where strict pollution emission restrictions were enforced alongside a comprehensive deployment of renewable resources. Remarkably, this scenario-maintained voltage deviations below 3 %, showcasing an exemplary model of what can be achieved with rigorous environmental standards and advanced integration strategies. The use of the HSGWO-MSOS algorithm in this scenario was pivotal, optimizing the placement and sizing of DG units to achieve an optimal balance between efficiency, reliability, and environmental impact. This led to a significant 30 % reduction in emissions and a 20 % improvement in system efficiency, effectively demonstrating how technology and policy can work together to enhance the sustainability and operational integrity of power systems. These results underscore the transformative potential of integrating advanced optimization algorithms with a strategic deployment of diverse energy resources, guided by strong environmental policies. The study not only offers a blueprint for future enhancements in power system management but also provides a compelling case for the role of technological innovation in achieving a sustainable energy future. As such, it serves as a critical reference point for utilities, policymakers, and researchers aiming to navigate the complex dynamics of modern energy systems towards more sustainable and resilient configurations.

CRediT authorship contribution statement

Xinghua Qi: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Bahadar Nawab Khattak:** Writing – review & editing, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Wenfu Liu:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Arif Alam:** Writing – review & editing, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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